

Project Development Phase

Cyber Security With AI

Project Name: AI-Enhanced Intrusion Detection System

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Model Performance Testing (VAPT)

Project team has filled the following information for VAPT testing for the target system.

S.No.	Parameter	Values
1.	Information Gathering	Footprinting – Target: Web application traffic classification (simulated network environment) Reconnaissance – CICIDS2017 dataset analyzed; 78 network flow features identified; attack patterns mapped for Brute Force, SQL Injection, and XSS
2.	Scanning the Target	Scanning Info – Network flow parameters scanned: Flow Duration, Total Fwd Packets, Total Bwd Packets, Total Length of Fwd Packets Risk Factors – High-risk: SQL Injection (payload length > 400 bytes), Brute Force (high packet count, long duration), XSS (moderate forward packets)
3.	Gaining Access	Access Process – REST API endpoint /predict receives JSON payloads; model classifies traffic in real-time Vulnerability Found – Web Attack vectors detected: Brute Force, SQL Injection, XSS; classified with ~98.9% accuracy
4.	Maintaining Access – Automation (AI Implementation)	AI Tools Used – Random Forest Classifier (Scikit-Learn), SMOTE (imbalanced-learn), Flask REST API Automation Implemented – Continuous packet flow monitoring via Live Threat Log; automated real-time classification pipeline; WAF integration via JSON API
5.	Covering Tracks & Report	Vulnerability Risk Factors – SQL Injection (Critical), Brute Force (High), XSS (Medium), BENIGN traffic (No risk) VAPT Report – Detection system achieves 98.9% accuracy; model optimized from 78 to 4 features; all 3 attack classes successfully detected in testing

Vulnerability Risk Summary

Attack Type	Risk Level	Detection Indicator
Web Attack – SQL Injection	Critical	Payload length > 400 bytes in forward packets
Web Attack – Brute Force	High	High packet count + long flow duration
Web Attack – XSS	Medium	Moderate forward packets, short duration
BENIGN Traffic	No Risk	Normal flow parameters, within baseline

AI Tools & Automation Stack

- ▶ Random Forest Classifier (Scikit-Learn) – Core ML model for traffic classification
- ▶ SMOTE (imbalanced-learn) – Handles class imbalance in training dataset
- ▶ Flask REST API – Receives JSON payloads and returns real-time predictions
- ▶ Live Threat Log – Continuous packet flow monitoring dashboard
- ▶ WAF Integration – Automated flagging via JSON API for Web Application Firewalls

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